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# Rapidly Learned Suppression of Salient Distractors via Multiple Regularities

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To stay focused, we must learn to direct attention to relevant information and ignore salient distractions. Many theories of attention suggest that the ability to ignore salient stimuli is driven by learning processes that require extensive exposure to distracting stimuli. The present study critically questions this assumption. Participants searched for a target and attempted to ignore a salient distractor. Importantly, a regularity in the salient distractor was held constant for short runs of five trials before changing. For example, the salient distractor could appear in a persistent color or location for five trials before changing. Shifts of gaze were used to determine if salient distractors captured attention or were instead suppressed. Across three experiments, we observed capture on the initial trial when the regularity changed, followed by immediate suppression on subsequent trials. Moreover, we demonstrate that this rapidly learned suppression occurs for multiple distractor regularities (featural, spatial, and combined). A fourth experiment showed that inserting an interruption trial produced capture, but suppression reemerged when the original regularity returned. Altogether, the results challenge theories proposing that salient stimuli cannot be ignored or that distractor suppression only results from slow statistical learning. Instead, the results support the view that the learned attentional control is highly adaptive, at least under certain conditions, in a manner that is broadly consistent with the Signal Suppression 2.0 account.

**Keywords:** attention, attentional capture, salience, suppression, learning

In everyday life, our visual system is constantly bombarded with salient stimuli that vie for our attention, such as flashy advertisements on a webpage or a brightly colored product on a supermarket shelf. Although salient stimuli are often assumed to attract attention, research shows that we learn to suppress them with experience. For instance, a flashing advertisement might initially draw attention, but repeated exposure helps us ignore it in favor of goal-relevant information. This learned suppression is typically assumed to be driven by statistical regularities such as when the salient stimulus appears in a consistent color or location (e.g., Gaspelin & Luck, 2018b, 2019; Savelson & Leber, 2024; Vatterott & Vecera, 2012; Wang & Theeuwes, 2018). Consequently, many theories of attention propose that suppression is learned through repeated encounters with salient stimuli.

The present study will investigate how flexibly attentional control can be implemented to resist distraction. Many theories propose that

attentional control is rather inflexible in that salient stimuli cannot be entirely ignored (Theeuwes, 2023) or that salient stimuli can only be ignored as a result of extensive learning (Wang & Theeuwes, 2018). The present study questions these assumptions. As will be seen, we find that attentional suppression can develop rapidly within one to two exposures to a distractor regularity and can integrate across multiple distractor regularities to improve suppression. These findings suggest that attentional suppression is a highly adaptive process that occurs as an immediate response to distraction by salient stimuli, as proposed by Signal Suppression 2.0 (Gaspelin et al., 2025).

## The Attentional Capture Debate

There has been substantial debate about whether salient stimuli can capture attention, regardless of our goals. Traditionally, the debate was divided into two theoretical accounts. Stimulus-driven accounts proposed that salient stimuli can involuntarily capture attention, even when completely task irrelevant (Theeuwes, 1992; Yantis & Jonides, 1984). For instance, Theeuwes (1992) had participants search for a target (e.g., a circle) among distractors (e.g., diamonds) and report the orientation of a line inside the target. On half of the trials, one object had a unique color compared with the other items (a color singleton), and this was never the target object. The presence of the color singleton slowed response times (RTs) relative to trials in which it was absent, which was taken as evidence that attention was captured by the color singleton due to its bottom-up salience and this capture slowed target detection.

By contrast, goal-driven accounts proposed that attention is only captured by salient stimuli matching an observer's goals (Folk et al., 1992). According to this account, an attentional template guides attention to visual objects that match the target features, and salient distractors can be ignored so long as they do not share features with the target. To explain evidence of stimulus-driven

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capture, Bacon and Egeth (1994) proposed that the task of Theeuwes (1992) encouraged a broad attentional template for salience (singleton-detection mode). When the target is salient (e.g., a lone diamond among many circles), participants might search for any salient object, which would enable capture by salient distractors. Importantly, when the task was adapted to prevent singleton-detection mode by including trials in which the target was not salient, capture effects were eliminated, suggesting that an observer's goals can powerfully influence attentional capture (see also Leber & Egeth, 2006).

### The Signal Suppression Account

To help resolve this debate, the Signal Suppression account offered a hybrid account of attentional capture (Sawaki & Luck, 2010). According to this account, salient stimuli do have an automatic power to attract attention, but that capture can be prevented via an inhibitory mechanism that is applied to salient distractors (see reviews by Gaspelin & Luck, 2018a, 2019). This account has been supported by several converging methodologies, such as event-related potentials (Drisdelle et al., 2023; Gaspar et al., 2016; Gaspelin et al., 2023; Gaspelin & Luck, 2018a; Sawaki & Luck, 2010; Stilwell et al., 2022; Tam et al., 2022), psychophysical approaches (Chang & Egeth, 2019, 2021; Gaspelin et al., 2015; Gaspelin & Luck, 2018b; Ma & Abrams, 2023; Won et al., 2022; H. Zhang et al., 2025), eye movements (Gaspelin et al., 2017; Stilwell et al., 2023; Won et al., 2019; Y. Zhang & Gaspelin, 2024), and single-unit recordings in monkeys (Cosman et al., 2018).

An important line of evidence for the signal suppression account has come from studies of shifts of gaze. Gaspelin et al. (2017) had participants search for a target shape (e.g., diamond) among distractor shapes, and a color singleton appeared at a nontarget location. Shifts of gaze were used to evaluate whether the color singleton captured attention or was instead suppressed. The results showed that eye movements were directed to the singleton distractor below the baseline level of the average nonsingleton distractor: an oculomotor suppression effect. This finding suggests that salient distractors can be suppressed to prevent capture from occurring and has since been replicated by several studies (Adams et al., 2023; Adams & Gaspelin, 2024; Drennan & Gaspelin, 2026; Gaspelin & Luck, 2018b, 2019; Hamblin-Frohman et al., 2022; Stilwell et al., 2023; Y. Zhang & Gaspelin, 2024).

Growing evidence indicates that distractor suppression is strongly shaped by prior experience (Anderson et al., 2021, 2026). Several studies have shown that distractor ignoring is improved when salient distractors repeatedly appear in the same location or with the same color (e.g., Adam et al., 2021; Adam & Serences, 2021; Anderson, 2026; Anderson et al., 2026; Gaspelin & Luck, 2018b, 2019; Hamblin-Frohman & Pratt, 2026; Ramgir & Lamy, 2022; Savelson et al., 2025; Savelson & Leber, 2024; Turatto et al., 2018; Vatterott & Vecera, 2012; Wang & Theeuwes, 2018). For example, in a seminal study, Vatterott and Vecera (2012) changed the color of a salient distractor every block of 48 trials. In the first half of the block, the salient distractor captured attention due to its new color. In the second half of the block, attentional capture effects were eliminated. Similarly, spatial suppression has been observed when salient distractors repeatedly appear in specific locations (e.g., Wang & Theeuwes, 2018).

To account for such findings, the signal suppression account has been updated to Version 2.0, which includes mechanisms to explain how distractor suppression is learned (Gaspelin et al., 2025). The full update includes a complete model of visual attention. The new model proposes that, after each attentional shift, the attended object is compared with a target representation (Yu et al., 2022, 2023). If the object matches the target representation, spatial and featural properties associated with the attended object are boosted. Alternatively, if the object mismatches the target representation, all spatial and featural properties associated with the attended object are downweighted. Thus, if a salient distractor captures attention, it will mismatch the target representation, and as a result, its featural and spatial properties will be downweighted in future shifts of attention. This relatively simple mechanism could explain how attentional tunings accumulate with experience to lead to suppression of salient distractors.

The Signal Suppression 2.0 account predicts that suppression should occur as a direct response to prior distraction. We have provided some evidence for this prediction (Y. Zhang & Gaspelin, 2026a). Participants performed an oculomotor suppression task, in which they searched for a target and attempted to ignore a color singleton. Importantly, the display colors were blocked for short runs of five trials, allowing us to measure rapid updates to attentional guidance. The results showed that, on the first trial in the color run, the newly colored singleton captured attention, as measured by shifts of gaze. On subsequent trials, the color singleton was immediately suppressed, after only a single exposure to the salient distractor. Additional experiments showed that reducing the strength of initial capture decreased the magnitude of suppression on subsequent trials, consistent with the Signal Suppression 2.0 account. These findings are consistent with the idea that learned suppression is preceded by an instance of capture. They also suggest that attentional suppression can develop rapidly within only one to two trials of experience, which is consistent with a mechanism that updates attentional control settings after each shift of attention (e.g., Adam et al., 2024; Gaspelin et al., 2019; Hamblin-Frohman & Pratt, 2026; Savelson et al., 2025).

In sum, the Signal Suppression 2.0 account predicts that salient objects will initially capture attention, which will cause attention to downweight the properties associated with the salient distractors on subsequent trials. Although we have provided some evidence for this account (Y. Zhang & Gaspelin, 2026a), additional evidence is needed to evaluate the breadth of this phenomenon and further test the Signal Suppression 2.0 account.

### How Rapid and Flexible Is Learned Suppression?

The present study will address two critical gaps in knowledge, which both relate to Signal Suppression 2.0's explanation of learned suppression. Here, we use the term "learned suppression" to refer to the suppression that develops as a function of experience across trials. Within the Signal Suppression 2.0 framework, this learning is assumed to be primarily implicit, although this is not intended to suggest that explicit processes never contribute. First, prior demonstrations of rapidly learned suppression have exclusively focused on suppression based upon featural regularities in color (Gaspelin et al., 2019; Hamblin-Frohman & Pratt, 2026; Savelson et al., 2025; Y. Zhang & Gaspelin, 2026a). It is therefore unclear whether rapidly learned suppression also occurs based upon other regularities. Specifically, there have been many demonstrations that

salient distractors can be suppressed based upon spatial regularities, such as when a distractor appears at a persistent location (e.g., De Waard et al., 2023; Ferrante et al., 2018; Wang & Theeuwes, 2018). However, most studies of spatial suppression suggest that distractor ignoring emerges relatively slowly because it is driven by associative processes that require extensive exposure to a distracting stimulus (e.g., 75–300 trials). This suggests that attentional control is slow to adapt to distraction, especially when compared with recent evidence that feature-based learning can be accomplished in one to two trials. If true, this would represent a significant limitation on learned suppression to reduce distraction.

Here, we distinguish between two accounts of learned suppression. Slow-adaptive learning accounts propose that learned suppression emerges relatively slowly (e.g., De Waard et al., 2023; Ferrante et al., 2018; Turatto, 2023; Turatto et al., 2018; Valsecchi & Turatto, 2021; Wang & Theeuwes, 2018). Examples of such accounts would include habituation accounts or statistical learning accounts that assume that learned suppression typically takes dozens, or even hundreds, of exposures to develop. Fast-adaptive learning accounts, however, suggest that learned suppression can develop much more rapidly, within a few exposures to a salient distractor (Adam & Serences, 2021; Gaspelin et al., 2019; Hamblin-Frohm & Pratt, 2026; Horstmann, 2015; Savelson et al., 2025; Y. Zhang & Gaspelin, 2026a). These accounts suggest a flexible attentional control mechanism that can quickly update to ignore salient stimuli. Although Signal Suppression 2.0 does not specifically propose that learning must always occur rapidly, it does suggest that attentional control settings are adjusted after each shift of attention, which should allow rapid learning to occur under certain conditions.

A challenge in resolving this discrepancy about the time course of learned suppression is that prior studies have differed in experimental paradigms, measures of attentional capture, and their assumption about what drives trial-by-trial changes in attentional capture, making the results difficult to compare directly. One issue is that rapid changes in capture across trials may reflect multiple short-term influences on attentional priority, including intertrial priming (Maljkovic & Nakayama, 1994, 1996) and surprise-induced attentional capture (Horstmann, 2002). For example, prior work on surprise-induced capture has shown that unexpected color singletons can capture attention when they violate observers' expectations about the upcoming display (Horstmann, 2002, 2005). Importantly, this capture is not simply a consequence of rarity or singleton status; it depends on whether the event mismatches explicit or implicit expectations (Horstmann, 2005), and rare singletons do not necessarily capture attention when they do not match the observer's attentional set (Horstmann & Ansorge, 2006). Simply put, there are many documented short-term biases in attention, and it is currently unclear how these short-term biases can accumulate to lead to more long-lasting biases (see Anderson et al., 2021; Gaspelin et al., 2025; Geng et al., 2019).

Another important issue concerns the experimental paradigms used to study learned suppression. Many studies supporting slow-adaptive learning accounts have used designs wherein the target and salient distractor shared a critical feature, which could prolong the time to reduce capture effects via contingent capture. For example, studies of spatial suppression commonly have participants search for a salient target while attempting to ignore a salient color singleton (Failing et al., 2019; Wang & Theeuwes, 2018). As reviewed above,

this type of search task may encourage goal-driven capture via singleton-detection mode (Bacon & Egeth, 1994), which might artificially prolong learned suppression: It might be difficult to learn to suppress a distractor that shares features with the target because suppression could sometimes yield suppression of the target itself. To estimate the maximal speed of learned suppression, it is necessary to evaluate suppression in a common paradigm that renders the salient distractor completely task irrelevant.

A second unresolved issue about learned suppression regards whether multiple regularities can be combined to improve suppression. The Signal Suppression 2.0 account proposes that all properties of the salient distractor will be downweighted in attentional control settings. This suggests that salient distractors should be more effectively suppressed when multiple regularities are present. Consistent with this interpretation, some prior studies have shown that adding a suppressible feature (e.g., color) to a salient distractor that is difficult to ignore (e.g., an abrupt onset) enables suppression of the salient distractor that was not originally possible (Adams et al., 2023; Adams & Gaspelin, 2024). This suggests that multiple sources of information can be integrated to improve distractor suppression. However, at present, there are no prior studies that have directly tested whether suppression is improved when both spatial and featural regularities are present beyond either regularity alone.

## The Present Study

The present study will examine learned suppression of salient distractors, with a specific emphasis on testing two predictions related to Signal Suppression 2.0. First, we will test whether distractor suppression is slowly learned or whether it can be rapidly learned. As reviewed above, prior studies have suggested that spatial suppression is slowly learned (e.g., 75–300 trials), which could reflect a limitation in attentional control, but many of these studies used methods in which the salient distractor was task relevant to some degree. We will therefore directly compare featural and spatial suppression using a common task established to produce rapid featural suppression in which the salient distractor is fully task irrelevant (e.g., Y. Zhang & Gaspelin, 2026a). Second, we will examine whether learned suppression is improved when both spatial and featural regularities are combined, as predicted by the Signal Suppression 2.0 account.

The present experiments used an oculomotor suppression task in which participants searched for a target and attempted to ignore a salient distractor. Importantly, a short-run technique was used in which a regularity in the salient distractor changed every five trials (Y. Zhang & Gaspelin, 2026a). In Experiments 1–3, we varied the type of regularity of the salient distractor to compare the relative strength and speed of learned suppression. In Experiment 1, a featural regularity in color changed every five trials. This experiment first established that rapidly learned suppression is possible, replicating prior results. Experiment 2 used the same basic approach, but a spatial regularity was manipulated in which the location of the salient distractor changed every five trials to test whether spatial regularities can also be rapidly learned. Experiment 3 then used a combined regularity in which both the location and color of the salient distractor changed every five trials, allowing us to test whether suppressive mechanisms integrate across multiple sources of available information. Finally, Experiment 4 examined how

lasting the rapidly learned suppression is in a version of the task that included an interruption trial in the middle of a run. This allowed us to test whether an interrupting stimulus would disrupt suppression when the previous regularity is reintroduced.

### Experiment 1: Featural Regularities

Experiment 1 was a control experiment examining whether salient distractors can be rapidly suppressed based upon featural regularities (see also Y. Zhang & Gaspelin, 2026a). Participants searched for a target shape (e.g., diamond) and attempted to ignore a color singleton (see Figure 1). Importantly, we used a short-run technique in which the display colors changed every five trials. Shifts of gaze were used to measure attentional capture and suppression as a function of experience with the salient distractor. Note that we have used a similar approach in a recent experiment to demonstrate rapidly learned suppression (Y. Zhang & Gaspelin, 2026a). The current control experiment aims to first replicate this result and will also serve as a crucial control experiment for comparison with other regularities in subsequent experiments.

There are competing predictions about attentional capture across the color run. Fast-adaptive learning accounts suggest that suppression is rapidly learned after only a few exposures. This should result in an initial instance of capture on the first trial after the color change (Trial 1) followed by suppression on later trials (Trials 2–5). Slow-adaptive learning accounts, however, predict that learned suppression will take many exposures to develop and therefore predict limited evidence of learned suppression within the brief color runs of five trials. Similarly, a purely stimulus-driven account would also predict that salient distractors will persistently capture attention based upon their bottom-up salience across the color run.

## Method

### Participants

Twenty-four students from the University of Missouri participated for payment of \$20 per hour or course credit (14 men and 10 women,  $M_{\text{age}} = 19.0$  years). The sample size was determined a priori based

upon effect sizes observed in a previous study using a similar task (Y. Zhang & Gaspelin, 2026a). Assuming the suppression effect is the same magnitude as the earliest reliable suppression observed in that study (Trial 2,  $d_z = 0.95$ ) and remains robust across subsequent trials, a minimum of 17 participants would be needed to obtain 95% power. We therefore collected 24 participants to err on the side of caution and to accommodate potential exclusions. As shown, we successfully detected a learned suppression effect in this experiment and replicated the effect in three subsequent experiments.

All participants had normal color vision as indicated by an Ishihara test and had normal or corrected-to-normal visual acuity as evaluated by a Snellen chart. The research protocol was approved by the internal review board at the University of Missouri.

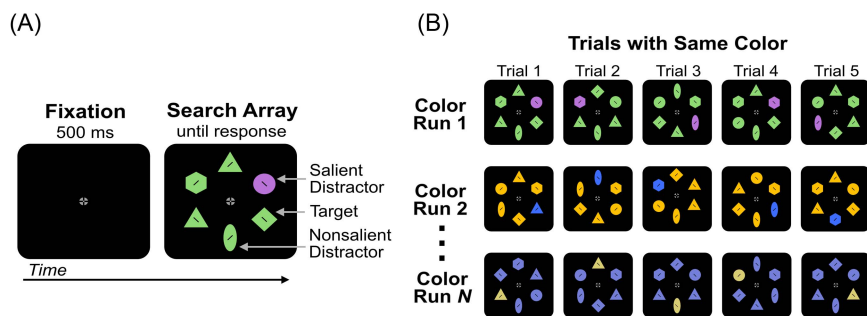
### Apparatus

Stimuli were presented using PsychToolbox for Linux on an Asus VG248QG LED monitor in a dimly lit room. A photosensor was used to measure the timing delay of the monitor (7 ms), and this was subtracted from all latency values in the article. The viewing distance was 100 cm and was controlled by a chinrest. An SR Research Eye Link 1000+ desk-mounted eye tracker measured gaze position monocularly from the right eye at 500 Hz.

### Stimuli and Procedure

As shown in Figure 1, an oculomotor suppression task was adapted to implement a short-run technique whereby the display colors changed every five trials (Y. Zhang & Gaspelin, 2026a). Each search array contained six shapes arranged in a notional circle at an eccentricity of  $4.3^\circ$ . The display included one diamond ( $1.7^\circ \times 1.7^\circ$ ) and one circle ( $1.7^\circ$  diameter). The remaining items were randomly selected from hexagons ( $1.7^\circ \times 1.7^\circ$ ), triangles ( $1.7^\circ$  height and  $1.7^\circ$  width), or ovals ( $2.0^\circ \times 1.4^\circ$ ), with the constraint that no more than two instances of a given shape appeared in a single search display. Each shape contained a small black line ( $0.2^\circ \times 0.1^\circ$ ), tilted  $45^\circ$  to the left or right. These lines were difficult to detect parafoveally, necessitating eye movements to perform the task. Each search display also included a centrally positioned, empirically optimized

**Figure 1**  
*Task and Stimuli for Experiment 1*



*Note.* (A) Trial progression for the task. Each trial began with a fixation display followed by a search array. Participants searched for a target shape (e.g., diamond) while attempting to ignore a salient distractor that was a color singleton. (B) A short-run technique was used to block the display colors that changed every five trials. Color values of the stimulus figures have been adjusted for illustrative purposes. See the online article for the color version of this figure.

fixation cross (Thaler et al., 2013) composed of a gray circle ( $30.2 \text{ cd/m}^2$ ,  $x = .290$ ,  $y = .294$ ,  $0.5^\circ$  diameter) with a black crosshair with a central gray dot measuring ( $0.1^\circ$  in diameter).

Each participant searched for a specific target shape (e.g., diamond) that remained constant for the entire experiment and was counterbalanced across participants. Heterogeneous distractor shapes were used to make the target nonsalient and prevent participants from searching broadly for any popout stimulus (Bacon & Egeth, 1994; Leber & Egeth, 2006), and this has been shown to be important for oculomotor suppression to develop (Gaspelin et al., 2017). On each trial, a salient distractor appeared as a uniquely colored object that differed from the colors of all other search objects. Participants were explicitly informed that the salient distractor would never be the target and were instructed to ignore it to the best of their ability.

As shown in Figure 1B, the display colors changed every five trials. To accomplish this, we generated a color wheel of 360 isoluminant color pairs using the International Commission on Illumination color space ( $L = 40$ ,  $R = 90$ ). Because all colors were selected from this same isoluminant color wheel, overall luminance was held constant across runs and only hue varied. For each color run, the target color was randomly selected from this color wheel, and the salient distractor color was  $180^\circ$  from the target color on the color wheel (i.e., the opposing color). This ensured that the color singleton was highly salient. The four nonsalient distractors always matched the target color. When the display colors changed, a new color pair was selected at random with the constraint that the new color pair differed by at least  $60^\circ$  from the previous color pair on the color wheel. This ensured that the new color pair was perceptually distinct from the previous color pair and was meant to reduce carryover effects of the attentional set between color runs.

Each trial began with the fixation cross. To initiate a trial, participants were required to maintain central fixation within a  $1.5^\circ$  radius around the fixation cross for 500 ms. Once this requirement was met, the search display was presented. Participants searched for the target shape (e.g., diamond) and reported the tilt direction (left or right) of the short line within the target. Manual responses were made via a speeded button press on a gamepad with the left- and right-shoulder buttons, respectively. The search array appeared until a response was made or until a 2,000-ms timeout. If the participants responded inaccurately or a timeout occurred, a 200-Hz tone and a prompt message showing either “wrong!” or “too slow!” (respectively) were displayed for 300 ms before the next trial began.

Performance feedback on mean accuracy and RT was provided at the end of each block. If a participant’s accuracy fell below 90% in a given block, a warning message was also displayed. Before the start of a new block, the eye-tracking system was recalibrated using a 9-point calibration. Recalibration was also triggered during the task if a participant failed to fixate the fixation cross for more than 8 s.

The experiment included 60 practice trials and 480 experimental trials, divided into nine blocks. The initial practice block did not contain any salient distractors and was simply meant to help participants learn the basics of the search task. Because each color run consisted of five trials, this resulted in 96 trials for each trial bin in the color run. This design ensures robust statistical power to study learned oculomotor suppression.

## Data Analysis

Eye movements were analyzed using a similar approach to previous studies using the oculomotor suppression task (Drennan & Gaspelin, 2026; Gaspelin et al., 2017; Y. Zhang & Gaspelin, 2024, 2026a). The onset of an eye movement was defined by exceeding both a minimum acceleration threshold of  $9,500^\circ/\text{sec}^2$  and a minimum velocity threshold of  $30^\circ$  per second. The first eye movement was categorized as the earliest saccade to be directed near a search object. Specifically, we defined an annulus centered on the fixation cross, with an inner radius of  $1.5^\circ$  and an outer radius of  $7.5^\circ$  from fixation. The first saccade landing within this annulus was identified as the first eye movement, and its destination was assigned to the nearest search. This approach effectively creates wedge-shaped interest areas around each display item (Leonard & Luck, 2011) and is meant to ensure that each eye movement analyzed falls within a reasonable distance of a search item. This approach was intended to minimize the classification of small deviations from central fixation as object-directed eye movements because such deviations may not reflect meaningful orienting toward a search item. In all experiments, only 3%–4% of first eye movements initially detected by the eye tracker landed outside this annulus. Saccadic latency was measured as the time to initiate the first eye movement from the onset of the search display.

Exclusion criteria were established based upon previous studies (Stilwell et al., 2023; Y. Zhang & Gaspelin, 2024, 2026a). The first block was considered practice and was excluded from analysis. Participants were replaced if manual RT, saccade latency, or manual error rates were more than 3.5 standard deviations from the group mean. No participants met these exclusion criteria. Trials were removed if participants did not make an eye movement, which was rare because the tilted lines were designed to be difficult to see from fixation (0.4%). Additionally, trials were removed if RTs were below 200 ms or above 2,000 ms (2.8%) or had abnormal saccade latencies (less than 50 ms or greater than 1,000 ms, 6.1%). Trials with incorrect manual responses (2.6%) were also excluded. In total, 10.0% of trials were excluded.

The primary data analyses were structured around the five-trial color runs. Trials within a color run were numbered using standardized Trials 1–5. Thus, Trial 1 is the initial trial in which the color changed to a new color. Trials 2–5 refer to the trials preceding this color change. As previously mentioned, 96 color runs were implemented in each experiment, and the salient distractor was always present, meaning there were 96 trials for each trial bin. This permitted strong statistical power to observe learned suppression of salient distractors.

All analyses of variance (ANOVAs) reported in this article include Greenhouse–Geisser corrected  $p$  values, and Cohen’s  $d$  for within-subject  $t$  tests is reported as  $d_z$ . For statistical comparisons with the short-run analyses, the false discovery rate (FDR) correction was applied to control for Type I errors with multiple comparisons (Benjamini & Hochberg, 1995). The FDR involves adjusting the criterion for significance rather than  $p$  value itself. We computed Jeffreys–Zellner–Siow Bayes factors ( $\text{BF}_{01}$ ; Rouder et al., 2009) to quantify evidence in favor of the null hypothesis for theoretically important null comparisons using JASP (Version 0.95.1; JASP Team, 2025).

## Transparency and Openness

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study, and we

follow Journal Article Reporting Standards (Appelbaum et al., 2018). All data, analysis code, and research materials are available on the Open Science Framework at <https://osf.io/94y5j/overview> (Y. Zhang & Gaspelin, 2026b). Data were analyzed using R, Version 4.5.0 (R Core Team, 2025), and the package ggplot, Version 3.5.2 (Wickham, 2016). This study's design and its analysis were not preregistered, and the data were collected in 2025–2026.

## Results

### Manual Responses

Table 1 shows manual RTs and error rates as a function of trial bin within a color run. As can be seen, manual RTs were slower on Trial 1 compared with Trials 2–5, which is consistent with the occurrence of capture on Trial 1 where the color changed. A one-way repeated measures ANOVA was conducted on manual RTs with the factor of trial bin (Trials 1–5). There was a significant main effect of trial bin,  $F(4, 92) = 14.70, p < .001, \eta_p^2 = .39$ . To characterize the time course of the RT reduction, we conducted planned adjacent-trial pairwise comparisons (Trials 1 vs. 2, Trials 2 vs. 3, Trials 3 vs. 4, and Trials 4 vs. 5). These pairwise comparisons confirmed that responses were significantly slower on Trial 1 than on Trial 2,  $t(23) = 5.80, p < .001, d_z = 1.18$ , whereas the remaining adjacent-trial comparisons were nonsignificant,  $ps > .10$ . This pattern indicates that the largest RT improvement occurred immediately after the first trial with relatively little additional change across later trials.

The same one-way repeated measures ANOVA was conducted on manual error rates. This analysis revealed no significant main effect of trial bin,  $F(4, 92) = 0.18, p = .947, \eta_p^2 = .01$ . This indicated that manual response errors were generally low, which is expected in this task.

### Destination of First Eye Movements

Figure 2A depicts heat maps of first eye movements as a function of trial bin (Trials 1–5). For simplicity, these heat maps depict a representative target-singleton angular distance of  $\pm 120^\circ$ . Displays were rotated so that the target always appeared at the top position, and the salient distractor location was indicated by a white arrow. On Trial 1, first eye movements were preferentially directed to the salient distractor (an oculomotor capture effect). By Trials 2–5, first eye movements were selectively directed away from the salient distractor (oculomotor suppression effect). This visualization clearly shows a rapidly learned suppression effect.

To formally analyze data, we collapsed across all potential angular distances. Figure 2B shows the percentage of first eye movements to

each type of search item (target, salient distractor, nonsalient distractor) as a function of trial bin within a color run (Trials 1–5). Here, the nonsalient distractor has been divided by four to provide a per-item estimate of the probability of being fixated to allow for comparison with the other search items. Oculomotor capture effects as a function of trial bin are shown in Figure 2C. This is a difference score of the percentage of first eye movements to the salient distractor minus the average nonsalient distractor (baseline). A positive value indicates an oculomotor capture effect above the nonsalient distractor baseline, and a negative value indicates an oculomotor suppression effect below the nonsalient distractor baseline. As shown, the salient distractor produced an initial oculomotor capture effect on Trial 1. By Trials 2–5, there was an oculomotor suppression effect whereby the salient distractor was suppressed.

A one-way within-subject ANOVA was conducted on oculomotor capture effects with the factor trial bin (Trials 1–5). There was a significant main effect of trial bin, indicating that there was a reduction in oculomotor capture across the trial bin,  $F(4, 92) = 45.72, p < .001, \eta_p^2 = .67$ . Preplanned one-sample  $t$  tests were conducted at each trial bin to examine whether the oculomotor capture and suppression significantly differed from zero. As explained above, significance was evaluated using the FDR procedure to control for multiple comparisons. Accordingly, the  $p$  values reported next are unadjusted, and significance was determined by comparing them against the corresponding FDR-adjusted critical  $p$  values. On Trial 1, the oculomotor capture effect was significantly greater than zero ( $+7.2\%$ ),  $t(23) = 7.25, p < .001, d_z = 1.48$ . This was immediately followed by oculomotor suppression below baseline levels of later trials, with significant oculomotor suppression observed as early as Trial 2 ( $-3.3\%$ ),  $t(23) = -3.40, p = .002, d_z = -0.70$ . This pattern of suppression continued across Trial 3 ( $-5.4\%$ ),  $t(23) = -4.91, p < .001, d_z = -1.00$ ; Trial 4 ( $-4.7\%$ ),  $t(23) = -4.46, p < .001, d_z = -0.91$ ; and Trial 5 ( $-5.5\%$ ),  $t(23) = -5.81, p < .001, d_z = -1.19$ .

We also examined the percentage of first eye movements to the target as a function of learning. A paired-samples  $t$  test showed that the percentage of first eye movements to the target was not significantly higher on Trial 5 than on Trial 1 ( $32.3\%$  vs.  $30.9\%$ ),  $t(23) = 1.16, p = .257, d_z = 0.24$ . Thus, target guidance was not necessarily improved by learning the color regularity, which would make sense as the target was defined by both color and shape. Thus, learning the color regularity would not necessarily lead to improved visual guidance to the specific target shape above those of other shapes in the display.

In sum, the results clearly indicate that suppression is learned within a single exposure to the salient distractor. These findings suggest that the learning mechanisms to reduce distraction may be much more efficient than previously assumed.

### Stability of Learned Suppression Effects

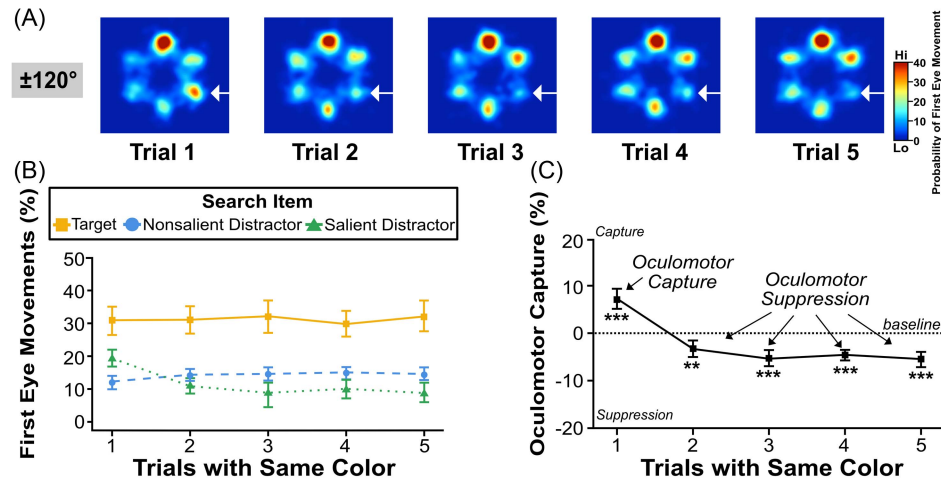
We were interested in how stable the learned suppression effects were across the experimental session. It is possible, for example, that changing the display colors would become less effective at inducing attentional capture as participants become accustomed to the regular change in display colors. This could lead to a reduction in the initial capture by salient distractors later in the experiment than the initial few color changes. We therefore compared oculomotor capture effects on Trial 1 between the first half of the experiment ( $+8.8\%$ ) and second half of the experiment ( $+5.8\%$ ), which showed no significant difference,  $t(23) = 1.20, p = .241, d_z = 0.25$ . A

**Table 1**

Manual Response Times (ms) and Error Rates Across Trial Bins in Experiments 1–3

| Trial bin | Experiment 1 | Experiment 2 | Experiment 3 |
|-----------|--------------|--------------|--------------|
| Trial 1   | 980 (2.3%)   | 893 (2.5%)   | 952 (1.9%)   |
| Trial 2   | 939 (2.3%)   | 870 (2.2%)   | 887 (1.9%)   |
| Trial 3   | 928 (2.5%)   | 858 (2.2%)   | 873 (2.7%)   |
| Trial 4   | 936 (2.2%)   | 851 (2.8%)   | 884 (2.7%)   |
| Trial 5   | 938 (2.1%)   | 854 (1.6%)   | 870 (2.1%)   |

Note. Error rates (%) are depicted in parentheses. Trial bin represents the trial number within a given color run of five trials.

**Figure 2***First Eye Movement Results for Experiment 1*

**Note.** (A) Heat maps of first eye movements as a function of trial bin in a representative angular distance ( $\pm 120^\circ$ ) between the target and salient distractor. The heat maps were generated so that the target was always at the top position, and the white arrow indicates the salient distractor position. All heat maps were plotted using the same color scale across trial bins. High (Hi) and Low (Lo) indicate the maximum and minimum likelihood that a first eye movement was directed to a given area of space and the color scale represents a probability density function of first eye movements after Gaussian smoothing. (B) First eye movements to each search item as a function of trial bin with the same color. (C) Oculomotor capture effects by salient distractor as a function of trial bin with the same color. Positive values indicate initial capture (Trial 1), while negative values reflect learned suppression (Trials 2–5). The downward trend illustrates the rapid shift from capture to suppression. For all plots, error bars represent within-subject 95% confidence intervals (Morey, 2008). See the online article for the color version of this figure.

\*\*  $p < .01$ . \*\*\*  $p < .001$ .

complementary Bayesian analysis provided anecdotal evidence in favor of the null hypothesis ( $BF_{01} = 2.44$ ), suggesting that the initial oculomotor capture effects were relatively stable and were not driven, for example, by an initial few exposures to the color change.

In addition, we also calculated learned suppression effects by comparing oculomotor capture effects on Trial 1 minus Trial 5. This gives an indication of how much oculomotor capture by the salient distractor was reduced across the trial bin. These learned suppression effects were also not significantly different in the first half of the experiment (13.5%) and the second half of the experiment (12.2%),  $t(23) = 0.44$ ,  $p = .665$ ,  $d_z = 0.09$ . A complementary Bayesian analysis provided moderate evidence for the null hypothesis ( $BF_{01} = 4.27$ ). This further suggests that learned suppression was fairly stable across the experimental session and was not driven, for instance, by the first few color runs in the experimental session.

### Saccadic Latency

A potential explanation of the learned suppression effect is that eye movements were slowed to prevent attentional capture from occurring. There is some reason to expect this could be a potential issue. For example, participants might have some awareness that attention was captured on Trial 1 and attempt to slow eye movements to avoid subsequent oculomotor capture. Indeed, there is evidence that participants have some awareness of oculomotor capture (Adams & Gaspelin, 2020, 2021), and saccadic slowing has been shown to improve top-down control in prior studies, albeit in different experimental paradigms than those of the present study

(e.g., Anderson & Mrkonja, 2021; Donk & Van Zoest, 2008). It is therefore important to rule out this explanation.

Table 2 depicts saccadic latency for each trial bin. Saccadic latency was calculated as the time to initiate the first eye movement after the onset of the search stimulus. If participants were slowing their eye movements to prevent capture across color runs, saccade latency should increase across the trial bin. As can be seen, we observe the opposite pattern predicted by a saccadic slowing account of the results. If anything, saccadic latencies were significantly faster on Trial 5 (192 ms) than on Trial 1 (199 ms),  $t(23) = -5.05$ ,  $p < .001$ ,  $d_z = -1.03$ . Thus, the learned suppression effect (i.e., the reduction in oculomotor capture across a color run) cannot be explained as slowing of eye movements to prevent capture from occurring. If anything, the opposite pattern was observed.

**Table 2**

*Saccadic Latency (ms) for Trial Bins With the Same Regularity in Experiments 1–3*

| Trial bin | Experiment 1 | Experiment 2 | Experiment 3 |
|-----------|--------------|--------------|--------------|
| Trial 1   | 199 (6.87)   | 195 (6.87)   | 196 (5.76)   |
| Trial 2   | 192 (6.46)   | 191 (7.14)   | 186 (4.91)   |
| Trial 3   | 193 (6.50)   | 190 (6.77)   | 187 (4.98)   |
| Trial 4   | 194 (6.71)   | 192 (7.29)   | 186 (4.98)   |
| Trial 5   | 192 (6.58)   | 191 (6.94)   | 187 (4.96)   |

**Note.** Standard errors of the mean (ms) are depicted in parentheses. Trial bin represents the trial number with the same regularity within a run of five trials.

## Discussion

Experiment 1 investigated learned suppression of salient distractors based upon featural regularities. Participants searched for a target and attempted to ignore a salient distractor. Importantly, the display colors changed every five trials, forcing participants to learn a new featural regularity to ignore the salient stimulus. This short-run technique allowed us to measure learned suppression to ascertain how flexible the attentional system is at adapting to new salient features. The findings showed that, on Trial 1, the salient distractor produced an oculomotor capture effect. By Trials 2–5, however, the features linked to the salient distractor were suppressed below baseline levels (see also Y. Zhang & Gaspelin, 2026a).

These findings are inconsistent with slow-adaptive learning accounts, which predict that attention requires many exposures to a salient distractor's regularity for suppression to occur. Similarly, a stimulus-driven account predicts that salient distractors should persistently capture attention regardless of prior experience, and this was also not observed. Instead, the results are consistent with fast-adaptive learning accounts, which predict that attentional control can be rapidly learned and implemented to ignore salient distractors. The general pattern is broadly consistent with the predictions of Signal Suppression 2.0: An event of attentional capture occurred, and this led to subsequent downweighting of the feature associated with the salient distractor. This control experiment will serve as a foundation for comparison of Experiments 2 and 3, which will examine learned suppression based upon other regularities.

## Experiment 2: Spatial Regularities

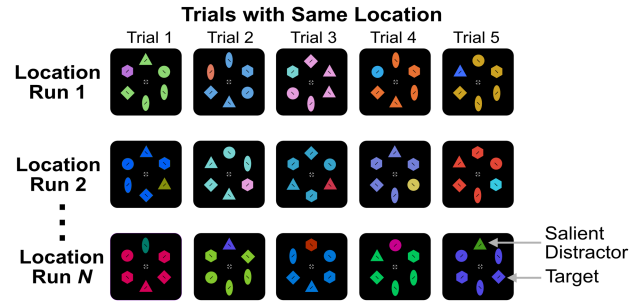
Experiment 1 demonstrated rapidly learned featural suppression. Prior studies have also shown evidence of spatial suppression, whereby salient distractors can be ignored when they appear at a frequent location (Failing et al., 2019; Leber et al., 2016; Wang & Theeuwes, 2018). However, spatial suppression is often presumed to be a result of slow statistical learning that develops across many exposures (e.g., 75–300 trials; De Waard et al., 2023; Valsecchi & Turatto, 2021; Wang & Theeuwes, 2018). There are many differences between these prior experiments and the approach of Experiment 1, which makes the results difficult to directly compare.

Experiment 2 therefore implemented the same general task and short-run technique but adapted it to investigate spatial suppression. The methods were identical to Experiment 1, except that the location of the salient distractor was held constant for short runs of five trials and then changed to a new location (Figure 3). To prevent feature-based suppression, the display colors changed every trial to a random color. This guaranteed that the salient distractor could only be suppressed by learning its spatial location.

The competing predictions are the same as Experiment 1. Slow-adaptive learning accounts predict that many repetitions are needed to learn the spatial regularities (e.g., De Waard et al., 2023; Valsecchi & Turatto, 2021; Wang & Theeuwes, 2018). According to these accounts, learned spatial suppression should not be possible within a five-trial run. Fast-adaptive learning accounts, however, predict that spatial regularities can be rapidly learned and that successful suppression should be instantiated within the five-trial runs (Y. Zhang & Gaspelin, 2026a). This result would also be broadly consistent with the Signal Suppression 2.0 account, which proposes an initial instance of capture will lead to immediate

**Figure 3**

*Task and Stimuli for Experiment 2*



*Note.* Experiment 2 used the same method as Experiment 1 but employed blocked runs of distractor location to examine spatial suppression. To prevent feature-based suppression, the colors were randomly changed on every trial. Color values of the stimuli have been adjusted for illustrative purposes. See the online article for the color version of this figure.

downweighting of all properties associated with the salient distractor, including its spatial location.

## Method

All methods were identical to Experiment 1, except for the following changes: A new sample of 24 participants was recruited from the University of Missouri for payment of \$20 per hour or course credit (15 men and nine women,  $M_{\text{age}} = 19.5$  years).

A similar short-run technique was used but with blocked runs of distractor location to observe learning suppression based upon spatial regularities (Figure 3). That is, every five trials, the location of the salient distractor changed to a new location that was randomly selected from the six locations. To prevent feature-based suppression, the colors were randomly changed on each trial. The selection of colors was random with the exception that the new color must be at least  $60^\circ$  away from the previous color. This was meant to ensure that the color change was sufficient in magnitude and reduce the possibility of carryover effect of the feature-based suppression from the previous trial.

The eye-tracking recording and analyses were the same as in Experiment 1. The first block was excluded as practice and did not contain a salient distractor. No participants were excluded for outlier performance (i.e., none were more than 3.5 SDs away from the group mean in terms of manual RT, manual error rates, or saccadic latency). Trials with manual RTs less than 200 ms or greater than 2,000 ms were removed (1.7%), along with those where participants did not move their eyes from central fixation (0.1%) or had abnormal saccade latencies (less than 50 ms or greater than 1,000 ms, 5.0%). Additionally, trials with incorrect responses (2.5%) were omitted. In total, 8.3% of trials were excluded from the final analysis.

## Results

### Manual Responses

Manual RTs and error rates are shown in Table 1. Manual RTs were slower on Trial 1 compared with later trials (Trials 2–5). A one-way repeated measures ANOVA on manual RT with the factor of trial bin showed a significant main effect of trial bin,  $F(4, 92) = 12.85$ ,  $p < .001$ ,  $\eta_p^2 = .36$ . Adjacent-trial pairwise comparisons between

Trials 1–2, Trials 2–3, Trials 3–4, and Trials 4–5 showed that responses were significantly slower on Trial 1 than on Trial 2,  $t(23) = 3.48$ ,  $p = .002$ ,  $d_z = 0.71$ , whereas the remaining adjacent-trial comparisons were not significant,  $ps > .10$ . This pattern indicates that the largest RT improvement occurred immediately after the first trial, with relatively little additional change across later trials.

The same analysis was repeated on manual error rates. Error rates were generally quite low (2.3%), and the analysis revealed no significant main effect of trial bin,  $F(4, 92) = 1.73$ ,  $p = .149$ ,  $\eta_p^2 = .07$ .

### Destination of First Eye Movements

Figure 4A shows heat maps of first eye movements as a function of trial bin using a representative target-salient distractor angular distance ( $\pm 120^\circ$ ). As can be seen, first eye movements were captured by the salient distractor on Trial 1, and the salient distractor was suppressed on the following Trials 2–5. To formally analyze this, Figure 4B collapses across all angular distances and depicts the percentage of first eye movements to each search item (target, salient distractor, nonsalient distractor) as a function of trial number within a run. As can be seen, there is evidence that the salient distractor was more likely to be fixated than the baseline level of the nonsalient distractor on Trial 1. This pattern is flipped by Trials 2–5, whereby the salient distractor is less likely to be fixated than the baseline level of the nonsalient distractor.

Figure 4C shows oculomotor capture effects as a function of trial bin. A one-way repeated measures ANOVA was conducted on oculomotor capture effects with the within-subject factor of trial bin. The analysis revealed a significant main effect of trial bin,  $F(4, 92) = 114.10$ ,  $p < .001$ ,  $\eta_p^2 = .83$ , indicating an overall decrease in capture from Trial 1 to Trial 5. Next, preplanned one-sample  $t$  tests were

conducted to assess whether the oculomotor capture significantly deviated from zero at each trial bin. A significant oculomotor capture effect was observed on Trial 1 (+8.8%),  $t(23) = 6.59$ ,  $p < .001$ ,  $d_z = 1.35$ , followed by oculomotor suppression on Trials 2–5: Trial 2 (–4.3%),  $t(23) = -4.53$ ,  $p < .001$ ,  $d_z = -0.92$ ; Trial 3 (–6.7%),  $t(23) = -6.75$ ,  $p < .001$ ,  $d_z = -1.38$ ; Trial 4 (–8.4%),  $t(23) = -8.54$ ,  $p < .001$ ,  $d_z = -1.74$ ; and Trial 5 (–10.2%),  $t(23) = -12.76$ ,  $p < .001$ ,  $d_z = -2.60$ . This suggests that spatial suppression can also be rapidly learned, just like feature-based suppression.

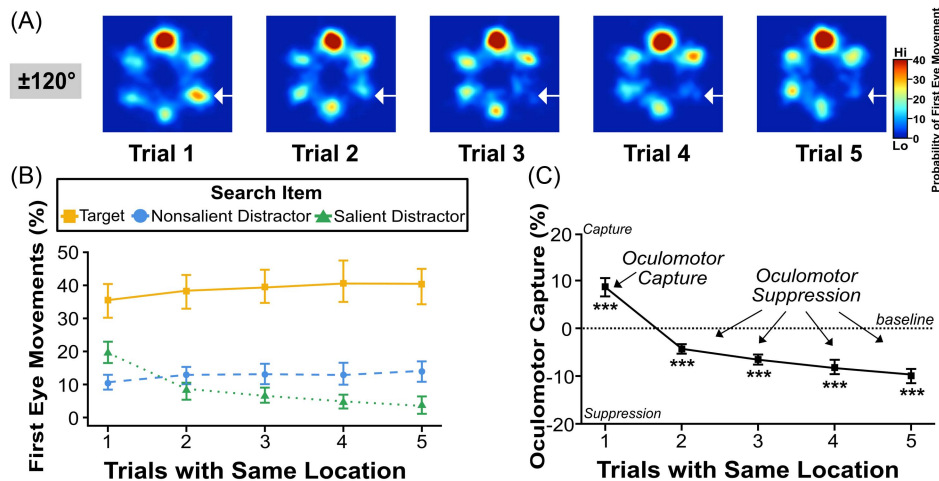
To further examine whether learned spatial suppression was accompanied by enhanced target guidance, we conducted a pre-planned paired-samples  $t$  test comparing first eye movements to the target on Trial 1 and Trial 5. The proportion of first eye movements to the target was significantly higher on Trial 5 than on Trial 1 (39.5% vs. 35.8%),  $t(23) = 2.45$ ,  $p = .023$ ,  $d_z = 0.50$ . Thus, learned spatial regularities may have contributed to improved direct guidance to the target in addition to reducing guidance to the salient distractor.

### Stability of Learned Suppression Effects

We again conducted an analysis to assess whether the initial capture effect decreased across the session. We first compared oculomotor capture effects on Trial 1 between the first and second half. The results revealed no significant difference between oculomotor capture effects in the first half (+10.7%) and the second half (+6.9%),  $t(23) = 1.69$ ,  $p = .104$ ,  $d_z = 0.35$ . A complementary Bayesian analysis provided anecdotal evidence in favor of the null hypothesis ( $BF_{01} = 1.35$ ). We also calculated learned suppression effects using oculomotor capture effects on Trial 1 minus Trial 5. These effects were also not significantly different in the first half (20.6%) and second half of the experiment (17.5%),  $t(23) = 1.14$ ,

**Figure 4**

*First Eye Movement Results for Experiment 2*



*Note.* (A) Heat maps of first eye movements as a function of trial bin for a representative angular distance between the target and salient distractor ( $\pm 120^\circ$ ). All heat maps were plotted using the same color scale across trial bins. High (Hi) and Low (Lo) indicate the maximum and minimum likelihood that a first eye movement was directed to a given area of space and the color scale represents a probability density function of first eye movements after Gaussian smoothing. (B) First eye movements to each search item as a function of trial bin with the same color and location of the salient distractor. (C) Oculomotor capture effects by salient distractors as a function of trial bin with the same color and location of the salient distractor. See the online article for the color version of this figure.

\*\*\*  $p < .001$ .

$p = .264$ ,  $d_z = 0.23$ . A complementary Bayesian analysis provided anecdotal evidence in favor of the null hypothesis ( $BF_{01} = 2.60$ ). This also suggests that pattern-learned suppression was relatively stable across the experimental session and was not entirely driven by the initial blocks of the experiment.

### Saccadic Latency

Table 2 depicts saccadic latency for each trial bin. As a reminder, a saccade slowing account would predict that learned suppression is accomplished by slowing saccades to improve top-down control within a trial run (i.e., from Trial 1 to Trial 5). For instance, after capture on Trial 1, participants might slow eye movements to attempt to implement top-down control. If anything, the opposite pattern was also observed here. Saccadic latencies were significantly slower on Trial 1 (195 ms) than Trial 5 (191 ms),  $t(23) = 2.39$ ,  $p = .026$ ,  $d_z = 0.49$ . Thus, the results cannot be explained by mere slowing of eye movements to prevent capture across a trial run.

### Discussion

Experiment 2 examined whether spatial regularities can also be rapidly learned and suppressed. Using the same paradigm and short-run technique as Experiment 1, the salient distractor appeared in a fixed location for five consecutive trials. The display colors randomly changed on each trial to prevent any feature-based suppression. The results showed that spatial suppression was rapidly learned within only one to two exposures, which is much faster than assumed by previous theories of statistical learning (e.g., De Waard et al., 2023; Valsecchi & Turatto, 2021; Wang & Theeuwes, 2018).

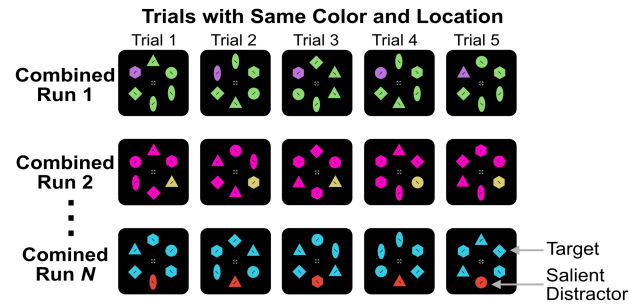
These findings are inconsistent with slow-adaptive learning accounts, which predict that attention is generally less flexible and takes many repetitions to learn to ignore a salient stimulus. Instead, the results support fast-adaptive learning accounts of distractor suppression, which propose a more adaptive attentional control system that can rapidly adapt based upon recent experience. Importantly, these results are broadly consistent with Signal Suppression 2.0's proposal that an initial instance of capture triggers rapid downweighting of properties associated with a salient distractor. This downweighting appears to operate not only over features (Experiment 1) but also over spatial location: Following initial capture, the distractor location is quickly downweighted, reducing subsequent priority and minimizing future capture.

### Experiment 3: Combined Regularities

The Signal Suppression 2.0 account predicts that all properties associated with a salient distractor should be suppressed (Gaspelin et al., 2025). A straightforward prediction is therefore that, when multiple regularities can be used to suppress a salient distractor, this should improve the strength of learned suppression. In other words, the attentional gain controls should simultaneously become tuned to ignore the spatial position and feature values of the salient distractor, leading to even more powerful suppression. Experiment 3 therefore tested whether combining both featural and spatial regularities would enhance suppression (Figure 5). In this experiment, the salient distractor changed in both color and location every five trials, allowing us to directly compare learned suppression for the

**Figure 5**

*Task and Stimuli for Experiment 3*



*Note.* Experiment 3 used the short-run method as Experiments 1 and 2 but combined both the spatial and featural regularities. Both the color and location of the salient distractor changed every five trials. Color values of the stimuli have been adjusted for illustrative purposes. See the online article for the color version of this figure.

combined property with the respective featural and spatial regularities alone from Experiments 1 and 2.

There are competing predictions about potential outcomes. According to the Signal Suppression 2.0 account, the combined regularities should be more strongly suppressed than either the featural or spatial regularity from Experiments 1 and 2. However, this might not be true. An alternative possibility is that one type of regularity will dominate attentional guidance and will be primarily used to suppress the salient distractor. For instance, spatial regularity may override featural information due to its direct mapping onto the priority map. In such a case, suppression of the combined regularities should be roughly equivalent to the featural or spatial regularity from the prior experiments.

### Method

All methods were identical to Experiments 1 and 2, except as follows: A new sample of 24 participants were recruited from the University of Missouri for payment of \$20 per hour or course credit (16 men and eight women,  $M_{age} = 18.8$  years).

A similar eye-tracking task was used as in Experiments 1 and 2. The primary difference was that the salient distractor's color and location were held constant for five trials before changing to a new color and location (Figure 5). That is, every five trials, the location of the salient distractor changed to a new location that was randomly selected from the six locations, and the color of the salient distractor changed to a new color (which was at least  $60^\circ$  away from the current color on the color wheel, as in Experiment 1).

The eye-tracking recording and analysis were the same as before. No participants were excluded for outlier performance (i.e., none were more than 3.5  $SDs$  away from the group mean in terms of manual RT, manual error rates, or saccadic latency). Trials with manual RTs less than 200 ms or greater than 2,000 ms (2.7%) were removed, along with those where participants did not move their eyes from central fixation (0.04%) or had abnormal saccade latencies (less than 50 ms or greater than 1,000 ms, 5.3%). Additionally, trials with incorrect responses (2.4%) were omitted. In total, 8.9% of trials were excluded from the final analysis.

## Results

### Manual Responses

Table 1 depicts manual RT and error rates as a function of trial bin. As shown, manual RTs were slower on Trial 1 compared with Trials 2–5. A one-way repeated measures ANOVA was conducted on manual RT with the factor of trial bin, and there was a significant main effect of trial bin,  $F(4, 92) = 39.06$ ,  $p < .001$ ,  $\eta_p^2 = .63$ . A follow-up pairwise  $t$  test confirmed that RTs were slower on Trial 1 than Trial 5,  $t(23) = 11.98$ ,  $p < .001$ ,  $d_z = 2.20$ . We next conducted adjacent-trial pairwise comparisons between trials (Trials 1 vs. 2, Trials 2 vs. 3, Trials 3 vs. 4, and Trials 4 vs. 5), which showed that responses were significantly slower on Trial 1 than on Trial 2,  $t(23) = 8.61$ ,  $p < .001$ ,  $d_z = 1.76$ , whereas the remaining adjacent-trial comparisons were not significant,  $ps > .10$ . This pattern indicates that the largest RT improvement occurred immediately after the first trial, with relatively little additional change across later trials.

In addition, the same one-way repeated measures ANOVA was conducted on manual error rates with the factor of trial bin. The analysis revealed no significant main effect of trial bin,  $F(4, 92) = 1.48$ ,  $p = .213$ ,  $\eta_p^2 = .06$ . This pattern indicates that participants consistently maintained a high level of accuracy throughout the experiment.

### Destination of First Eye Movements

We first confirmed that a similar pattern of results was observed as in Experiments 1 and 2, whereby there was oculomotor capture on Trial 1 followed by oculomotor suppression on Trials 2–5. Figure 6A shows heat maps of first eye movements as a function of trial bin using a representative target-salient distractor angular distance

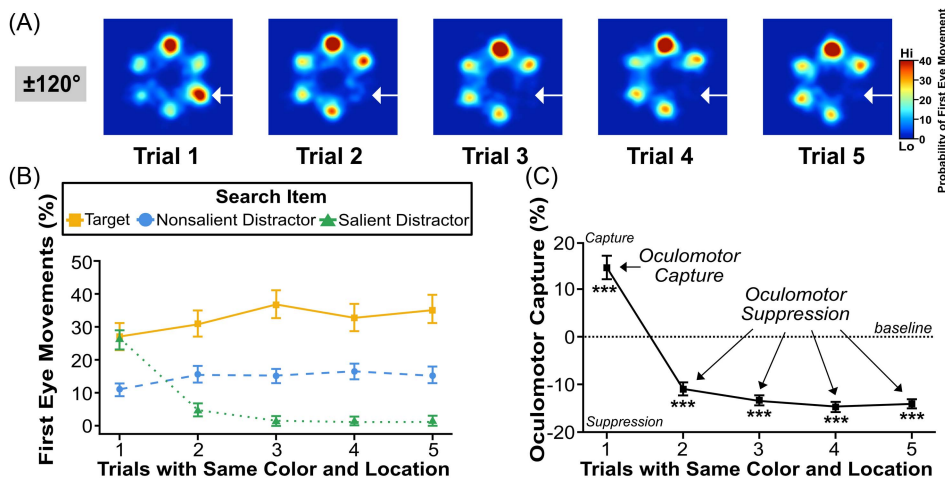
( $\pm 120^\circ$ ). As shown, we observed a pattern similar to the previous experiments: There was evidence that the salient distractor was enhanced above the baseline level of nonsalient distractors on Trial 1 (capture), and this pattern reversed on Trials 2–5 (suppression). Importantly, the initial capture and learned suppression were both much more powerful in the current task than in previous experiments.

Figure 6B depicts the percentage of first eye movements to each search item for each trial bin, pooled across every potential angular distance. Figure 6C shows oculomotor suppression effects for each trial bin. A one-way repeated measures ANOVA was conducted on oculomotor capture effects with the within-subject factor of trial bin. The analysis revealed a significant main effect of trial bin,  $F(4, 92) = 292.76$ ,  $p < .001$ ,  $\eta_p^2 = .93$ , indicating an overall decrease in capture from Trial 1 to Trial 5. Next, a series of preplanned one-sample  $t$  tests were conducted to assess whether the oculomotor capture significantly deviated from zero across trial bins. A significant initial capture effect was observed on Trial 1 (+14.7%),  $t(23) = 10.40$ ,  $p < .001$ ,  $d_z = 2.12$ , followed by rapidly learned suppression on Trials 2–5: Trial 2 (–11.1%),  $t(23) = -13.09$ ,  $p < .001$ ,  $d_z = -2.67$ ; Trial 3 (–13.4%),  $t(23) = -21.32$ ,  $p < .001$ ,  $d_z = -4.35$ ; Trial 4 (–15.0%),  $t(23) = -19.88$ ,  $p < .001$ ,  $d_z = -4.06$ ; and Trial 5 (–14.2%),  $t(23) = -21.47$ ,  $p < .001$ ,  $d_z = -4.38$ .

To further examine whether learned suppression with combined regularity was accompanied by enhanced target guidance, we conducted a preplanned paired-samples  $t$  test comparing target-directed first eye movements on Trial 1 and Trial 5. The proportion of first eye movements to the target was significantly higher on Trial 5 than on Trial 1 (35.6% vs. 27.4%),  $t(23) = 4.40$ ,  $p < .001$ ,  $d_z = 0.90$ . Thus, the combined regularity may have also improved guidance to the target in addition to reducing orienting to the salient distractor.

**Figure 6**

*First Eye Movement Results for Experiment 3*



*Note.* (A) Heat maps of first eye movements as a function of trial bin for a representative angular distance between the target and salient distractor ( $\pm 120^\circ$ ). All heat maps were plotted using the same color scale across trial bins. High (Hi) and Low (Lo) indicate the maximum and minimum likelihood that a first eye movement was directed to a given area of space and the color scale represents a probability density function of first eye movements after Gaussian smoothing. (B) First eye movements to each search item as a function of trial bin with the same color and location of the salient distractor. (C) Oculomotor capture effects by salient distractors as a function of trial bin with the same color and location of the salient distractor. See the online article for the color version of this figure.

\*\*\* $p < .001$ .

### Featural, Spatial, Versus Combined Regularities

This experiment allowed us to test whether the combined regularities (Experiment 3) would improve learned suppression compared with the featural regularity (Experiment 1) or spatial regularity (Experiment 2) alone. If attention is guided by a single regularity, the combined regularity should produce similar capture and suppression to both lone regularity conditions from the previous experiments. Alternatively, if multiple regularities are used to guide attention, then we should see improved suppression beyond either individual regularity alone.

Figure 7 depicts oculomotor capture effects as a function of trial bin for each regularity type (i.e., featural, spatial, combined). As shown, the combined regularities resulted in stronger initial capture on Trial 1 than either featural or spatial regularities alone. Importantly, oculomotor suppression effects on Trials 2–5 were also much stronger in combined runs than for either featural or spatial regularities. This suggests that the attentional system learned to use both regularities to suppress salient distractors, in a manner consistent with the Signal Suppression 2.0 account.

To formally analyze this, we conducted a mixed-design ANOVA on oculomotor suppression effects with the within-subject factor of trial bin (Trials 1–5) and the between-subject factor of regularity type (featural, spatial, combined). The analysis revealed a significant main effect of trial bin,  $F(4, 276) = 385.33, p < .001, \eta_p^2 = .85$ , indicating that all experiments showed a learned suppression effect whereby oculomotor capture by the salient distractor was reduced across a trial run. There was also a significant main effect of the regularity type,  $F(2, 69) = 14.71, p < .001, \eta_p^2 = .30$ . Importantly, there was a significant interaction between trial bin and regularity type,  $F(8, 276) = 26.48, p < .001, \eta_p^2 = .43$ , indicating that the degree of learned suppression differed between regularity types.

The key question was whether oculomotor suppression was enhanced for the combined regularities compared with the featural or spatial regularity alone. Preplanned between-subject  $t$  tests compared oculomotor capture effects for each regularity type at each trial bin. These revealed that on Trial 1, the combined regularities (Experiment 3) exhibited significantly greater initial oculomotor capture than either

the featural regularity (Experiment 1),  $t(46) = 4.34, p < .001, d = 1.25$ , or the spatial regularity (Experiment 2),  $t(46) = 3.07, p = .004, d = 0.89$ . Additionally, for Trials 2–5, the combined regularities (Experiment 3) exhibited significantly stronger oculomotor suppression than the featural regularity (Experiment 1) on Trial 2,  $t(46) = 5.97, p < .001, d = 1.72$ ; Trial 3,  $t(46) = 6.38, p < .001, d = 1.84$ ; Trial 4,  $t(46) = 7.90, p < .001, d = 2.28$ ; and Trial 5,  $t(46) = 7.45, p < .001, d = 2.15$ , and the spatial regularity (Experiment 2) on Trial 2,  $t(46) = 5.30, p < .001, d = 1.53$ ; Trial 3,  $t(46) = 5.78, p < .001, d = 1.67$ ; Trial 4,  $t(46) = 5.35, p < .001, d = 1.55$ ; and Trial 5,  $t(46) = 3.83, p < .001, d = 1.11$ . A post hoc power analysis was conducted on these between-subject comparisons using the average effect size ( $d = 1.60$ ), which revealed that the average power ( $1 - \beta$ ) was 0.99 [0.86–1.00], suggesting that the present comparisons were sufficiently powered. These results demonstrate that combining spatial and featural regularities not only increases initial attentional capture but also substantially enhances the subsequent oculomotor suppression effect.

Altogether, these findings suggest that the combined regularities are more effectively suppressed than spatial or featural regularity alone. At a theoretical level, this suggests that both forms of attentional guidance are effectively integrated to enable the salient distractors to be suppressed, consistent with the revised signal suppression account.

### Discussion

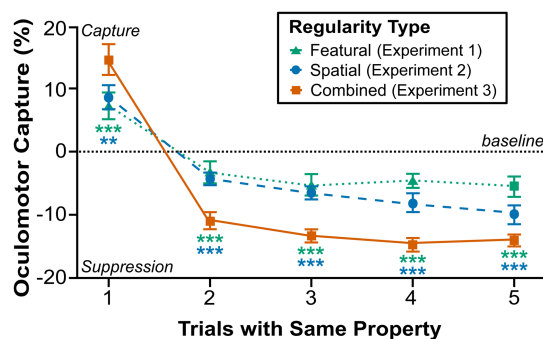
Experiment 3 examined whether combining spatial and featural regularities enhances learned suppression. According to the Signal Suppression 2.0 account (Gaspelin et al., 2025), all properties associated with a salient distractor are downweighted when it mismatches the target. This account therefore predicts stronger suppression when multiple regularities are present. An alternative possibility is that one regularity might dominate learned suppression. If so, suppression in Experiment 3 should resemble that in Experiments 1 or 2. The results clearly showed that combining regularities led to significantly stronger initial capture and improved learned suppression, suggesting that the attentional system flexibly integrates multiple cues to rapidly reduce distraction. One possibility is that changing both the color and location of the salient distractor may have created a stronger attentional priority signal than changing either dimension alone, and this stronger initial orienting may have provided a stronger learning signal for subsequent suppression (Y. Zhang & Gaspelin, 2026a).

### Experiment 4: Interrupted Regularities

Experiments 1–3 showed rapidly learned suppression of multiple regularities. However, it remains unclear whether these learning effects reflect a more long-term form of learning that can survive intervening visual experience or whether they depend primarily on immediate repetitions of the distractor regularity from the previous trial. Experiment 4 was designed to examine this issue by testing whether the learned suppression could survive an interruption from an intervening stimulus display.

Experiment 4 used runs of combined regularities, which previously produced the strongest suppression effects. The trial runs were extended to eight trials to enable a new task design that used two types of runs. On *pure* runs, the color and location of the salient distractor remained constant for the entire eight-trial run before changing. These runs served as a control condition. On *interrupted*

**Figure 7**  
Learned Suppression Effects for Regularity Types in Experiments 1–3



*Note.* Oculomotor capture effects for each regularity type by trial bin. Asterisks represent between-subject  $t$  test comparisons for combined versus featural (green) and combined versus spatial (blue). See the online article for the color version of this figure.

\*\*  $p < .01$ . \*\*\*  $p < .001$ .

runs, the singleton appeared in the same color and location for Trials 1–4, but an interruption trial occurred on Trial 5, whereby both the color and location of the salient distractor changed. After the interruption, the original regularity from Trials 1–4 returned for the remainder of the run (Trials 6–8). This new condition allowed us to test whether suppression of the original distractor regularity would survive an intervening stimulus or whether capture would return following the interruption.

There are competing predictions about performance postinterruption trial (Trial 6) in which the previous regularity is restored. If learned suppression is based upon the immediate regularity from the previous trial, there should be oculomotor capture on the post-interruption trial (Trial 6). In other words, the postinterruption trial involves a change in the regularity from the previous trial. Thus, the interruption trial might completely disrupt learned suppression on the postinterruption trial, leading to capture by the salient stimulus. Alternatively, if learned suppression reflects the accumulation of priority across multiple trials, then some level of suppression of the original regularity should return on the postinterruption trial, even if it is somewhat weakened relative to the pure condition. That is, there should be some recovery of suppression of the previous regularity without evidence of capture.

## Method

All methods were identical to Experiment 3, except as follows: A new sample of 24 participants was recruited from the University of Missouri for payment of \$20 per hour or course credit (eight men and 16 women,  $M_{\text{age}} = 22.3$  years).

Trial runs were extended to eight trials, whereby the feature and location of the salient distractor remained constant (i.e., the combined regularity of Experiment 3). Two types of runs were used: *pure* and *interrupted*. On pure runs, the color and location of the salient distractor remained constant for the eight-trial run before changing to a new combined regularity. On interrupted runs, the same combined regularity was initially maintained (Trials 1–4), but an interruption trial was inserted on Trial 5, in which both the display colors and distractor location changed. In addition, the salient distractor appeared at a new (randomly selected) location. After the interruption trial, the original regularity from Trials 1–4 returned for the remainder of the run (Trials 6–8). These two conditions were randomly selected throughout the experiment.

The experiment included 96 practice trials and 864 experimental trials, divided into 10 blocks. The initial practice block did not contain any salient distractors and was simply meant to help participants learn the basics of the search task. Because each trial run consisted of eight trials, this resulted in 108 trial runs across the experiment. This design ensured robust statistical power to study learned oculomotor suppression.

The eye-tracking recording and analysis are the same as before. No participants were excluded for outlier performance (i.e., none were more than 3.5 *SDs* away from the group mean in terms of manual RT, manual error rates, or saccadic latency). Trials with manual RTs less than 200 ms or greater than 2,000 ms (1.5%) were removed, along with those where participants did not move their eyes from central fixation (0.07%) or had abnormal saccade latencies (less than 50 ms or greater than 1,000 ms, 4.8%). Additionally, trials with incorrect responses (2.4%) were omitted. In total, 7.8% of trials were excluded from the final analysis.

## Results

### Manual Responses

We had no direct predictions about manual responses but analyzed them for the sake of completeness. Manual RTs were slower on Trial 1 compared with later trials (Trials 2–8) in both pure and interrupted runs (see Table 3). A two-way repeated measures ANOVA was conducted on manual RT with two within-subject factors: trial bin (Trials 1–8) and run type (pure and interrupted). There was a significant main effect of trial bin,  $F(7, 161) = 29.87, p < .001, \eta_p^2 = .56$ , indicating that manual RTs varied systematically across trials, with faster responses in later trials. The main effect of run type was marginally significant,  $F(1, 23) = 3.72, p = .066, \eta_p^2 = .14$ . Importantly, there was a significant interaction between trial bin and run type,  $F(7, 161) = 3.88, p < .001, \eta_p^2 = .14$ , indicating that the pattern of manual RTs across the run differed between the pure and interrupted conditions.

In addition, the same two-way repeated measures ANOVA was conducted on manual error rates with the factors of trial bin and run type. The analysis did not reveal a significant main effect of trial bin,  $F(7, 161) = 1.02, p = .419, \eta_p^2 = .04$ . Likewise, there was no significant main effect of run type,  $F(1, 23) = 1.78, p = .195, \eta_p^2 = .07$ . Moreover, there was no significant interaction between trial bin and run type,  $F(7, 161) = 0.79, p = .599, \eta_p^2 = .03$ . Overall, these results indicate that participants consistently maintained a high level of accuracy throughout the experiment.

### Destination of First Eye Movements

The purpose of this experiment was to determine whether learned suppression could survive an interrupting stimulus, at least to some degree. Figure 8A depicts oculomotor capture effects for each trial bin (Trials 1–8) and run type (pure and interrupted). The interruption trial occurred on Trial 5 in the interruption condition. As shown, the interruption caused a clear attentional capture effect on Trial 5, as expected by all theories. The key question is whether this interrupting stimulus would completely disrupt previously learned suppression when the previous regularity returned on Trials 6–8. As shown, suppression below a baseline level returned on Trial 6, even after the interrupting stimulus. If the interruption trial completely erased previous learning, capture should have occurred on Trial 6, at a similar level to Trials 1 and 5. This was clearly not

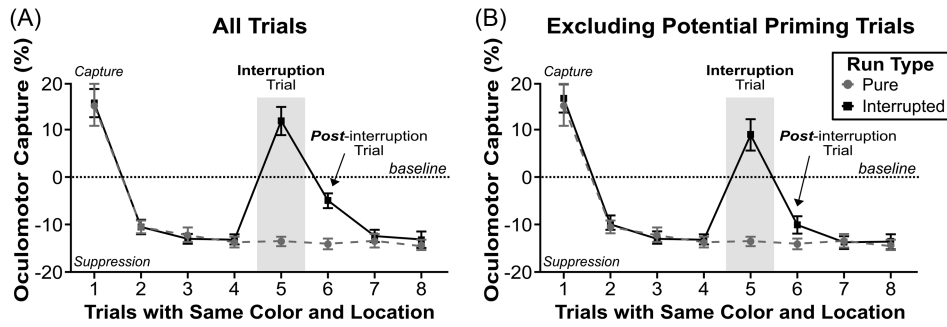
**Table 3**

*Manual Response Times (ms) and Error Rates Across Trial Bins in Experiment 4*

| Trial bin | Experiment 4 |             |
|-----------|--------------|-------------|
|           | Pure         | Interrupted |
| Trial 1   | 965 (1.0%)   | 972 (2.3%)  |
| Trial 2   | 906 (2.0%)   | 906 (2.4%)  |
| Trial 3   | 894 (2.3%)   | 898 (2.6%)  |
| Trial 4   | 895 (2.1%)   | 889 (2.7%)  |
| Trial 5   | 892 (1.8%)   | 944 (2.0%)  |
| Trial 6   | 893 (2.5%)   | 910 (2.3%)  |
| Trial 7   | 894 (2.1%)   | 900 (2.4%)  |
| Trial 8   | 897 (2.1%)   | 892 (1.9%)  |

*Note.* Error rates (%) are depicted in parentheses. Trial bin represents the trial number within a given color run of eight trials.

**Figure 8**  
*Oculomotor Capture Effects for Experiment 4*



*Note.* Oculomotor capture effects by salient distractors as a function of trial bin with the same color and location of the salient distractor. Separate panels depict results from all trials (A) and results after excluding trials in which the interruption trial could have produced potential priming of the salient distractor (B). Positive values indicate capture by the salient distractor, whereas negative values indicate suppression relative to the nonsalient distractor baseline level. The gray-shaded region marks Trial 5, on which a new intervening stimulus was introduced in the interrupted runs. The arrow marks Trial 6, the first postinterruption trial, when the original regularity returned.

observed. This indicates that some degree of learned suppression from Trials 1–4 could survive an interrupting stimulus.

A two-way repeated measures ANOVA was conducted on oculomotor capture effects with the within-subject factor between trial bin (Trials 1–8) and run type conditions (pure vs. interrupted). The analysis revealed a significant main effect of trial bin,  $F(7, 161) = 56.40, p < .001, \eta_p^2 = .95$ . The main effect of run type was significant,  $F(1, 23) = 141.65, p < .001, \eta_p^2 = .86$ . Crucially, a significant interaction between trial bin and run type was observed,  $F(7, 161) = 44.37, p < .001, \eta_p^2 = .93$ , indicating that the time course of oculomotor capture differed between the pure and interrupted conditions.

Next, a series of preplanned one-sample  $t$  tests was conducted to assess whether the oculomotor capture significantly deviated from zero across trial bins in each run type condition. In the pure condition, a significant oculomotor capture effect was observed on Trial 1 (+15.8%),  $t(23) = 6.72, p < .001, d_z = 1.37$ , followed by oculomotor suppression on Trials 2–8: Trial 2 (−10.6%),  $t(23) = -14.14, p < .001, d_z = -2.89$ ; Trial 3 (−12.3%),  $t(23) = -13.65, p < .001, d_z = -2.79$ ; Trial 4 (−13.7%),  $t(23) = -22.05, p < .001, d_z = -4.50$ ; Trial 5 (−13.4%),  $t(23) = -19.88, p < .001, d_z = -4.06$ ; Trial 6 (−14.0%),  $t(23) = -15.90, p < .001, d_z = -3.25$ ; Trial 7 (−13.6%),  $t(23) = -16.49, p < .001, d_z = -3.37$ ; and Trial 8 (−14.5%),  $t(23) = -17.22, p < .001, d_z = -3.52$ . In the interrupted condition, there was also a significant capture effect on Trial 1 (+16.0%),  $t(23) = 10.71, p < .001, d_z = 2.19$ , followed by significant suppression on Trials 2–4: Trial 2 (−10.6%),  $t(23) = -12.14, p < .001, d_z = -2.48$ ; Trial 3 (−13.1%),  $t(23) = -20.20, p < .001, d_z = -4.12$ ; and Trial 4 (−13.3%),  $t(23) = -16.19, p < .001, d_z = -3.31$ . Crucially, capture reemerged on Trial 5 when the interrupting color-location regularity appeared (+12.4%):  $t(23) = 6.51, p < .001, d_z = 1.33$ . When the original regularity returned, we again observed significant oculomotor suppression on Trials 6–8: Trial 6 (−5.1%),  $t(23) = -6.99, p < .001, d_z = -1.43$ ; Trial 7 (−12.5%),  $t(23) = -14.54, p < .001, d_z = -2.97$ ; and Trial 8 (−13.4%),  $t(23) = -15.55, p < .001, d_z = -3.18$ .

For interrupted runs, we analyzed the postinterruption trial (Trial 6) with key comparison points in other conditions via within-subject

$t$  tests. We first compared Trial 6 in interrupted runs with Trials 1 and 5 in interrupted runs (which also introduced a new regularity). These analyses revealed a significantly lower capture effect on Trial 6 than Trial 1 (−5.1% vs. +16.0%),  $t(23) = -12.38, p < .001, d_z = -0.21$ , or Trial 5 (−5.1% vs. +12.4%),  $t(23) = -10.68, p < .001, d_z = -0.18$ . This confirms that Trial 6 performance showed enhanced suppression compared with other trials in which a new regularity was introduced, confirming that some level of learned suppression remained for the old regularity. We also compared Trial 6 between the interrupted and pure runs, which revealed that suppression was significantly weaker on Trial 6 in the interrupted runs than in the pure runs (−5.1% vs. −14.0%),  $t(23) = 9.51, p < .001, d_z = 1.94$ . This indicates that although significant below-baseline suppression remained on Trial 6 following an interruption, it was not as strong as suppression without an interruption.

In sum, the results suggest that learned suppression returned for the previous regularity after an intervening stimulus was introduced, at least to some degree. This suggests that the observed learning effects are not merely a result of expectations based upon the previous-trial regularity and that learned suppression reflects a more long-term learning mechanism that accumulates across multiple trials, as suggested by Signal Suppression 2.0.

### **Excluding Potential Priming Trials**

We also conducted an exploratory analysis that removed any influence of intertrial priming from the interruption trial on the postinterruption trial. Due to randomization, the target color and target location on the interruption trial (Trial 5) could sometimes directly prime the singleton distractor's color or location on the postinterruption trial (Trial 6). For example, on a small subset of trials, the singleton distractor on the postinterruption trial (Trial 6) might appear in the same color or location as the target from the interruption trial (Trial 5). This immediate priming could have artificially boosted the number of eye movements to the salient distractor on Trial 6 (Maljkovic & Nakayama, 1994, 1996; Talcott et al., 2022; Talcott & Gaspelin, 2020; Toledano et al., 2024).

We therefore repeated the analysis from the previous section but excluded entire runs that had any potential priming from interruption trials (Figure 8B). Specifically, a run was excluded if the target location on Trial 5 matched the singleton location on Trial 6 or if the target color on Trial 5 was within  $\pm 30^\circ$  of the singleton color on Trial 6. As shown, eliminating these trials clearly improved the strength of suppression on the postinterruption trial. Within-subject  $t$  tests confirmed that oculomotor suppression was significantly larger on Trial 6 in the priming-excluded data set ( $-10.4\%$ ) than in the full data set ( $-5.1\%$ ),  $t(23) = 5.70$ ,  $p < .001$ ,  $d_z = 1.16$ . As before, oculomotor suppression was significantly below zero on the post-interruption trial ( $-10.4\%$ ),  $t(23) = -11.76$ ,  $p < .001$ ,  $d_z = -2.40$ , indicating that some level of suppression survived the interruption trial. Interestingly, the suppression on Trial 6 was still somewhat weaker than in the pure condition ( $-14.0\%$  vs.  $-10.4\%$ ),  $t(23) = -4.24$ ,  $p < .001$ ,  $d_z = -0.86$ .

In sum, if we exclude trial runs where the postinterruption distractor (Trial 6) is directly primed by the interruption trial (Trial 5), we observe even stronger recovery of the previously learned regularity. This further suggests that the learned suppression effects can survive an intervening stimulus, suggesting preservation of the previously learned suppression of the regularity.

## Discussion

Experiment 4 tested whether rapidly learned suppression depends primarily on uninterrupted repetitions of the same distractor regularity. The results showed that introducing a different distractor regularity produced renewed capture on the interruption trial, indicating that the attentional system remained sensitive to changes in the currently salient distractor. However, when the original regularity returned on Trial 6, it did not appear to be treated as a new regularity requiring complete relearning (i.e., there was no capture). Instead, suppression of the original regularity persisted, although it was reduced relative to uninterrupted repetitions in the pure condition. This effect was even stronger if we removed interruptions that directly primed the singleton distractor on the postinterruption trial. Altogether, the results suggest that these learned suppression effects are not merely a result of immediate expectations based upon the prior trial, and instead suppression appears to reflect flexible-but-persistent adjustment to attentional priority that can survive brief intervening experience. This is directly consistent with the type of accumulation of priority across learning instances suggested by the Signal Suppression 2.0 account (Gaspelin et al., 2025).

## General Discussion

Understanding how distraction is prevented is essential for advancing theories of attention. Much research has suggested that the attentional system learns to suppress salient distractors with experience (e.g., Gaspelin & Luck, 2018b, 2019; Savelson et al., 2025; Savelson & Leber, 2024; Vatterott & Vecera, 2012; Wang & Theeuwes, 2018). However, critical gaps remain in understanding how the attentional system learns to suppress salient distractors. Signal Suppression 2.0 offers a mechanistic explanation by proposing that an initial instance of capture triggers immediate adjustments to attentional priority, downweighting properties associated with a distractor to reduce subsequent capture (Gaspelin et al., 2025). This account suggests that the attentional system should rapidly learn to

ignore salient distractors, at least under certain conditions. This account also suggests that the attentional system can flexibly integrate across multiple regularities to improve distractor suppression.

The present study aimed to directly test these predictions across a series of experiments. We used a short-run technique to investigate how flexibly attentional templates can be updated to enable distractor suppression (Y. Zhang & Gaspelin, 2026a). Participants performed an attentional capture task adapted for eye tracking, in which they searched for a target and attempted to ignore a salient distractor. Across three experiments, a critical property of the salient distractor was held constant for five trials before changing. We manipulated whether this regularity was featural, spatial, or combined. This short-run approach allowed us to measure rapidly learned suppression with great precision by averaging across the trials within a run (Reinhart & Woodman, 2015). Shifts of gaze were used to determine if salient distractors captured attention or were suppressed.

Experiment 1 was a control experiment examining whether salient distractors can be rapidly suppressed based upon featural regularities (e.g., color), replicating prior results (Y. Zhang & Gaspelin, 2026a). Participants completed a search task wherein distractor colors remained constant for five trials. On Trial 1 (the initial trial with color change), the newly colored distractor captured attention as measured by eye movements. On Trials 2–5, the salient distractor was suppressed below baseline levels. These results indicate that feature-based suppression can be rapidly learned within only a few trials.

Experiment 2 investigated whether spatial regularities of salient distractors could also be rapidly learned. The same short-run technique was used except that the location of the salient distractor changed every five trials. Feature-based suppression was prevented using random display colors on each trial. The results mirrored those of Experiment 1. Shifts of gaze were initially captured on Trial 1, followed by immediate suppression on Trials 2–5. These results suggest that spatial suppression can be acquired on a much shorter timescale (e.g., 1–2 trials) than assumed by slow-adaptive learning accounts that emphasize gradual accumulation across repeated exposures (e.g., De Waard et al., 2023; Wang & Theeuwes, 2018). Instead, the results are consistent with Signal Suppression 2.0's proposal that initial capture rapidly recalibrates attentional priority, downweighting the distractor location to minimize subsequent capture.

Experiment 3 examined whether combining multiple regularities could improve learned suppression beyond either regularity alone. Both the color and location of the salient distractor changed every five trials. If both regularities can be used to suppress the salient distractors, suppression should be improved compared to experiments with only one regularity to suppress. If only a single regularity can be used to suppress the distractor, then suppression should be similar to prior experiments. The results showed that the combined regularities resulted in stronger learned suppression than either the featural regularity or the spatial regularity alone. Although our between-subjects design does not allow for a test of strict additivity, it did reveal that the combined regularity is suppressed to a greater degree than either regularity alone, suggesting that attentional guidance was not merely guided by one regularity or another. This pattern aligns with Signal Suppression 2.0, which posits that initial capture functions as a learning signal that rapidly downweights properties associated with a salient distractor across multiple

dimensions (Gaspelin et al., 2025). When both feature (e.g., color) and spatial regularities were present within a run, the system could downweight both dimensions, yielding stronger suppression than when only a single regularity was available.

Experiment 4 provided a further test of whether learned suppression could survive an interruption from an intervening stimulus, at least to some degree. In this experiment, trial runs were sometimes interrupted by a search display in which the salient distractor appeared in a new color and location. The results clearly showed that below-baseline suppression returned on the postinterruption trials when the previous regularity was reintroduced. Further analyses showed that when we eliminated the influence of intertrial priming from the interrupting stimulus, this returned suppression was even stronger on the postinterruption trial. These findings suggest that the rapidly learned suppression effects in these experiments are not merely the product of immediate expectations based upon the previous trial and instead reflect a more long-term mechanism that accumulates across multiple trials.

Altogether, the current findings advance the understanding of attention in three important ways. First, the results indicate that learned suppression of distracting stimuli can develop much more rapidly than is typically emphasized by statistical learning accounts, which propose that learned suppression develops gradually and requires extensive repetition of a salient distractor's features or locations (e.g., Wang & Theeuwes, 2018). Instead, learned suppression can occur after only one or two exposures to a salient distractor, consistent with fast-adaptive learning accounts highlighting a highly flexible attentional control mechanism. Second, the results show that learned suppression is not confined to a single regularity: When multiple regularities were available, they were integrated to produce stronger suppression than either regularity alone. Third, the interruption design of Experiment 4 showed that learned suppression is the result of accumulated priority values based upon prior experience, and this learning cannot be entirely eliminated by a single intervening stimulus. Together, these findings align with Signal Suppression 2.0, which proposes that initial capture rapidly recalibrates attentional priority by downweighting all properties associated with a salient distractor when it mismatches the target, thereby reducing the likelihood of subsequent capture (Gaspelin et al., 2025; Y. Zhang & Gaspelin, 2026a).

Although the current results highlight that learned suppression can occur rapidly under certain conditions, we do not claim that learned suppression is always rapid. Instead, we suggest that how quickly suppression is learned will vary across tasks and other environmental demands. For example, we suspect that learned suppression might be prolonged when salient distractors are task relevant and match potential target features (e.g., Turatto et al., 2018; Wang & Theeuwes, 2018). In such cases, it may be difficult for participants to learn to downweight the salient distractor because this could also involve (accidentally) downweighting properties associated with the target or reverting to simplified strategies that allow rapid detection of a target (e.g., singleton-detection mode). Future research is therefore needed to determine the role that task relevance plays in the speed of learned suppression, preferably in controlled designs that allow one to directly compare suppression of relevant and irrelevant salient distractors. The speed of learned suppression might also be influenced by other design factors, such as the frequency and predictability with which the distractor appears (Horstmann, 2015; Won et al., 2019), the certainty of the distractor's

location, and internal awareness that distraction is occurring (Adams & Gaspelin, 2020, 2021). Understanding whether these factors influence the speed of learned suppression could help refine existing models of attentional capture and learned attentional control, more generally.

There are potential parallels between the current findings and the literature on attentional capture by unexpected salient distractors. As reviewed above, this literature has shown that salient distractors, which are unexpected and rare, can powerfully capture attention (Adam et al., 2024; Horstmann, 2002, 2015) and that distractor suppression is reduced for infrequent salient distractors (Won et al., 2019). The key difference between this literature and the current findings is that the salient distractors in the present study were present on every trial but were accompanied by an infrequent change in properties (e.g., that occurred on only 20% of trials in Experiments 1–3). Future work is therefore needed to better understand how expectations about a salient distractor's presence and expectations about a distractor's properties might influence attentional capture. The Signal Suppression 2.0 account proposes that attentional priority accumulates as a function of prior experience with the features and locations of a salient distractor. However, it makes no clear distinction between the presence of a salient distractor and how the properties of a salient distractor might influence the accumulation of experience-dependent suppression. This might therefore be a fruitful avenue for future work.

More generally, task demands may influence the extent to which changes in attentional guidance reflect reduced distractor guidance, enhanced target guidance, or both. Consistent with this possibility, target-directed first eye movements were not improved across the trial runs in Experiment 1 but were significantly higher in Experiments 2 and 3. Thus, spatial and combined regularities may have improved direct target guidance, although enhanced target guidance cannot fully account for the below-baseline suppression of the salient distractor relative to nonsalient distractors. This issue is especially important when below-baseline effects are defined relative to nonsalient distractors because changes in target guidance may influence the proportion of first eye movements directed to all distractor items, including the nonsalient distractor baseline (e.g., see Chang & Egeth, 2019; Hamblin-Frohman et al., 2022; McDonald et al., 2026; Oxner et al., 2023). Such designs could also help distinguish whether changes in performance reflect reduced distractor guidance, enhanced target guidance, or both.

The present findings are consistent with a largely implicit form of learned suppression; however, we cannot completely rule out the possibility that explicit processes also contributed under some conditions. For instance, when a distractor regularity remained stable within five-trial runs, it is possible that participants learned to intentionally attend to the color singleton on the initial change trial to learn the new regularity. However, this possible strategy does not provide a complete explanation of the present results for a few reasons. First, we found that the learned suppression effects were relatively stable across the experiment in Experiments 1–3. Second, and perhaps more directly, robust attentional capture was observed on the critical interruption trial in Experiment 4. These interruption trials provided no information about the upcoming color regularity, and there should therefore have been no incentive to voluntarily attend the singleton on these trials. That being said, future work is needed to more directly examine how implicit and explicit processes

shape learned suppression, as there are well-documented instances whereby explicit knowledge can produce suppression (Arita et al., 2012; Carlisle, 2023).

The present findings have important implications for other areas of cognitive psychology. First, although the present findings mostly concerned attentional capture, they could reflect a general mechanism by which visual attention adapts to recent experience rather than a process that is specific to salient distractors. Much prior work has shown that observers can also learn to suppress distractors that are not salient (e.g., Addleman & Störmer, 2023; Pang et al., 2024; Prakash & Hollingworth, 2026; Stilwell & Vecera, 2019). From this perspective, learned suppression may reflect a broader attentional mechanism that is used to filter distracting information. However, further work is needed to understand whether learned suppression of salient stimuli and nonsalient stimuli relies on similar learning mechanisms.

Learned suppression could also be used to reduce the ability of salient stimuli to interfere with later stages of cognition such as working memory. There has been a long history of suggesting that attention and other gating processes can be used to help control access to working memory (Cowan et al., 2024; Gaspar et al., 2016; Jones et al., 2024; Vogel et al., 2005). Some recent findings have suggested that salient items may enjoy an automatic benefit of representation in visual working memory (Constant & Liesefeld, 2021, 2023). However, other research has suggested that attention can be used to mitigate the role of salience on working memory representations (Gaspelin & Cowan, 2025, 2026). Future research may more definitively link inhibitory processes involved in reducing distraction to control over working memory representations.

In conclusion, the present study indicates that learned suppression of salient distractors can occur rapidly, within only one or two trials. We also found that an initial instance of capture occurs before learned suppression and that combining multiple regularities leads to enhanced learned suppression. These results suggest a highly flexible attentional mechanism that can be quickly adapted to ignore distracting stimuli, which is consistent with the Signal Suppression 2.0 account.

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